Roll No. | | | | | | | |

Paper Code: TMC 401

Paper Name COMPUTER GRAPHICS AND VISUAL COMPUTING

Time: 3 Hours MM: 100

Each question has three parts (a,b,c) . Attempt any two parts of each question

Q1. **(10 X 2 = 20 Marks)**

a. Explain parallel Projection? How perspective projection is different from parallel projection? What is the importance of vanishing point in projection?

b. Calculate the coordinates of a given unit cube having a point A, at center (0,0,0) rotated about z axis by 90 degree anti clock wise. Show the transformation of the rotated cube.

c. What do you understand by back face detection. Explain Z-buffer algorithm.

Q2. **(10 X 2 = 20 Marks)**

a. What is ROTATIONAL transformation? Explain types of transformations.

b. For a given triangle of coordinates A (10,30) B(30,30) C(20,15) find the following
   1. Reflection about X-axis
   2. Scaling about point A

c. Explain Bresenham circle drawing algorithm? Why circle is divided in octants in circle drawing algorithms?

Q3. **(10 X 2 = 20 Marks)**

a. Digitize the pixel points using DDA algorithm for a line segment A(20,12) B(31,19) using DDA line drawing algorithms

b. Write short note on any two
   1. Shearing Transformation
   2. Oblique Projection
   3. Principle of animation

c. What is inside-outside test? Explain odd-even parity and winding number rule for checking if a point is inside or outside a given area.

Q4. **(10 X 2 = 20 Marks)**

a. Derive bresenham line algorithm? What are the limitations of bresenham algorithm.

b. Explain 4-connected and 8-connected model for filling. Explain boundary fill algorithm.

c. Explain the working of LCD and LED. How raster scan is different from random scan?

Q5. **(10 X 2 = 20 Marks)**

a. Describe the Sutherland Hodgeman polygon clipping algorithm. What are convex and concave polygons?

b. Consider two raster systems with the resolutions of 640x480, 1280x1024. What size frame buffer (in KB) is needed for each of these systems to store 12 bits/pixel , if 10 seconds video with 30 fps is loaded.

c. Briefly explain the Cohen-Sutherland line clipping algorithm?

Roll No. | | | | | | | |

Paper Code: TMA 201

B.Tech.
Back End Examination 2022
Second *Semester*

Engineering Mathematics II

Time: 3:00 Hours

MM: 100

Note:
(i) This question paper contains five questions.

Each question contains three parts a, b & c. Attempt any two parts of choice from each question.

Q1. (10 X 2 = 20 Marks)

a. Solve the differential equation of $(D^2+D+1)y=\cos 2x$.

b. Solve $\frac{d^2y}{dx^2}+y=\tan x$ using the method of variation of parameters.

c. Solve $(x^2D^2+4xD+2)y=e^x$.

Q2. (10 X 2 = 20 Marks)

a. Find the Laplace transform of $\sin 2t\cos 3t$.

b. Find Laplace inverse of $\frac{1}{s^2-7s+12}$.

c. Using the Convolution Theorem find $L^{-1}\left\{\frac{s}{(s^2+1)(s^2+4)}\right\}$.

Q3. (10 X 2 = 20 Marks)

a. Express $J_5(x)$ in terms of $J_0(x)$ and $J_1(x)$.

b. Show that $\int_{-1}^{1} x^2 P_{n-1}(x)P_{n+1}(x)\,dx=\frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}$.

c. Prove that $P_n(x)=\frac{1}{n!2^n}\frac{d^n}{dx^n}\left(x^2-1\right)^n$.

Roll No. | | | | | | | |

End Semester Back Paper Examination-2022

Name of the Program: B.Tech (CSE) *Semester: IV*

Paper Name: Computer Organization *Course Code: TCS-404*

Time: 3 Hour's **Maximum Marks: 100**

Note:

(i) All questions are compulsory.
(ii) Answer any two questions among a,b,c in each question
(iii) Total marks in each question are 20 marks
(iv) Each question carries 10 marks

Q.1	**(10x2=20 Marks)**	
a)	Compare The Von Neumann and Non Von Neumann Model.	CO-1
b)	Perform signed multiplication (+12) and (-8) using Booth's Multiplication Algorithm.	
c)	Perform division of unsigned integer 12/3 using restoring division algorithm.	
Q.2	**(10x2=20 Marks)**	
a)	Differentiate between the Hardwired CU and the Micro-programmed CU?	CO-2
b)	What is instruction format? Explain instruction cycle with the help of flow chart.	
c)	What are Shift Micro operations? Starting from initial value of R= 100010110 determine the sequence of binary values in R after a Logical shift left, followed by a circular shift right, followed by an arithmetic shift Left and circular shift right..	
Q.3	**(10x2=20 Marks)**	
a)	What are Interrupts? Briefly Explain various types of interrupts.	CO-3
b)	Explain IOP Processor with the help of block diagram	
c)	Define I/O Interface. What are the requirements of it? Explain its structure	
Q.4	**(10x2=20 Marks)**	
a)	What is Memory Hierarchy; Show the memory hierarchy with the help of a diagram showing speed, cost and size.	CO-4
b)	Write short notes on: I. RAM & ROM	

	ii. EPROM iii. Cache Memory iv. Virtual memory	
c)	Explain LRU cache replacement policy with a suitable explains.	
Q.5	(10x2=20 Marks)	
a)	What is pipeline? Explain instruction pipelining with the help of an example.	CO-5
b)	A non pipelined system takes 60 ns to process a task. The same task is processed via 5 Stage pipeline having their delays as 3 ns, 2ns, 5ns, 15ns, 3ns. The buffer registers with delay of 1 ns each are used in between the stages. What is the speedup achieved for 60 tasks?	
c)	Differentiate between RISC & CISC architecture in detail	

-----------------------------------THE END-----------------------------------

Roll No. [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Paper Code: TCS-201

Even End Sem Back Paper Examination December 2022

Course - B.Tech *Semester – 2nd*

Subject Name: Programming for Problem Solving

Subject Code: TCS 201

Time: Three Hours MM: 100

Note:

i. This question paper contains five questions.
ii. All Questions are compulsory.
iii. Instructions on how to attempt a question are mentioned against it.
iv. Total marks assigned to each questions are twenty.

Q1. (Attempt any two questions of choice from a, b and c). (2*10=20 Marks)

a.

i. Design a C program to accept N real numbers from the user, compute the average and then store the numbers those are above the average in another array called **aboveAvg** and below average in the array **belowAvg**. Display the average and content of both the arrays. Write the corresponding algorithm for the same.

ii. Describe multi-dimensional arrays with an illustration. List the advantages of Arrays in real world applications.

b.

i. Define an Array. Explain & illustrate the significance of **Row Major Order** method of storage of array elements.

ii. Implement a C program by accepting an integer through the command line arguments and compute its factorial. Display the computed factorial to the output screen.

c. Write a C program to check that inputted string is palindrome or not. (without using string library function)

Q2. (Attempt any two questions of choice from a, b and c). (2*10=20 Marks)

a.

i. Explain the storage of strings in C by emphasizing on the significance of **null character**. Describe the working of **strncmp** and **strncpy** functions with an example.

ii. Accept a sentence from the user and capitalize the first letter of each of the words of the sentence. Assume the words are separated uniformly by a single space. Implement using a C program. Draw a flowchart for the same.

b.

i. "***Usage of dynamic memory allocation by a programmer leads to more efficient usage of memory than a static memory allocation***". Justify the statement with an illustration.

ii. Design a non-recursive C function that will test whether a pattern string is the prefix of a text string or not. Accept the text string and the pattern string in the calling program and display an appropriate message to the output screen if the pattern is present or not present.

Example:
"ABC" is a prefix of **"ABC Company"**, but **"AABC"** is not.

c. Write a program to accept name, age and basic salary of 5 employee and calculate the total salary of the employee as-
Total sal =Basic +TA+DA where TA is 15% of Basic Salary, DA is 5% of Basic Salary. Display the information of the employees in the ascending order of their total sal.

Q3. (Attempt any two questions of choice from a, b and c). (2*10=20 Marks)

a. Define the following terms with an example for each: **2.5 x 4 = 10 M**

i. dangling pointer
ii. null pointer
iii. wild pointer
iv. segmentation fault

b.

i. **"Pointers can be used alternatively to the access the arrays"**. Justify & demonstrate using a C program.
ii. Implement a C program to accept numbers using pointers. Design a function Sort that sorts and returns the sorted numbers in descending order using pointers. Display the sorted numbers in the main program.

c. What is dereferencing of pointer variable? Explain how these expression will work assuming that p is a integer pointer and holding address 300-

1. x=*p++	2. p=p+2
3. x=*(p+2)	4. x=++*p

Q4. (Attempt any two questions of choice from **a, b and c**). (2*10=20 Marks)

a.

i. Illustrate different ways of declaring a structure. A certain grocery store would like to store ten different categories of items with their quantities and prices. Write a structure declaration for the same in C.
ii. Design a program in C that numbers each of the lines in a file. Accept N lines from the user and write to a file **"numbered.txt"** by pre-fixing a line number followed by a space to each of sentences of the file.
For example: 1 First Sentence.......
2 Second Sentence ....
3 Third Sentence,.................. & so on till
:

N Nth Sentence.

b.

i. Elaborate on working of files using **fscanf** and **fprintf** functions. Explain with a snippet of C code.

ii. A Nursery plantation house would like to automate to maintain different saplings breeds based on their name, number of saplings and the price per sapling. Write a C program using appropriate data types to store the details of N saplings. Display the total cost of maintaining the Nursery to the output screen

Sample Input:
For N=2
Enter the name of the sapling: **Red Rose**
Enter the quantity: 200
Enter the price per sapling: 50
Enter the name of the sapling: **Hibiscus**
Enter the quantity: 100
Enter the price per sapling: 35

Total Cost of maintaining the Nursery (Rs): **13500**

c. Write a C program to read records from a file and calculate grade of each student and display it.
Grade - A if marks>=75
Grade - B if marks>=60
Grade - C if marks>=50
FAIL if marks>=0

Q5. (Attempt any two questions of choice **from a, b and c**). (2*10=20 Marks)

a.

i. Describe the working of **range** function with a sample code illustrating the same. **3M**

ii. Write a Python code to accept a list of integers from the user and then display the same by sorting it in descending order. **4M**

iii. List the differences between list and a tuple. **3M**

b.

i. Dictionary in Python is more flexible data structure that allows storing a pair of items. Support the statement with an example.

ii. Design a program in Python to accept & store the name of items with their respective quantities to a dictionary. Then eliminate duplicate items from the dictionary and display it to the console. Write the corresponding algorithm for the same.

c. Find Output for the following code-

(Consider the following code for a 16 bit compiler. Ignore the punctuation error. if any and header files)

<table>
<tr>
<td>

i)int main() [2]

```
{
   char str[]={'H','E','L','P','\0','\n','M','E','\0', '\0'};
   printf("%s\n",str);
   printf("%d  %d\n",strlen(str),sizeof(str));
   return 0;
}
```

</td>
<td>

ii)int sum(int); [4]

```
int main()
{
        printf( " %d ", sum(146) );
        printf( " %d ", sum(681) );
        return 0;
}
int sum(int n)
{
        static int s;
        while(n!=0)
        {
                s=s+n%10;
                n=n/10;
        }
        return sum;
}
```

</td>
</tr>
<tr>
<td>

iii)int main() [2]

```
{
 enum value={val1=0,val2,val3,val4,val5}var;
printf("%d",sizeof(var));
return 0;
}
```

</td>
<td>

iv)int main() [2]

```
{
 struct emp
 {
  char n[20];
  int age;
 };
struct emp e1={"Dravid",23};
struct emp e2=e1;
if(e1==e2)
printf("The structure variable are equal ");
}
```

</td>
</tr>
</table>

Roll No. | | | | | | | | **Paper Code:** TCS-408

B.Tech End Semester (Back) Examination, Dec,2022

4th *Semester*

Paper Name : **Java Programming**

Time: Three Hours | MM: 100

Note:

(i) This question paper contains five questions.

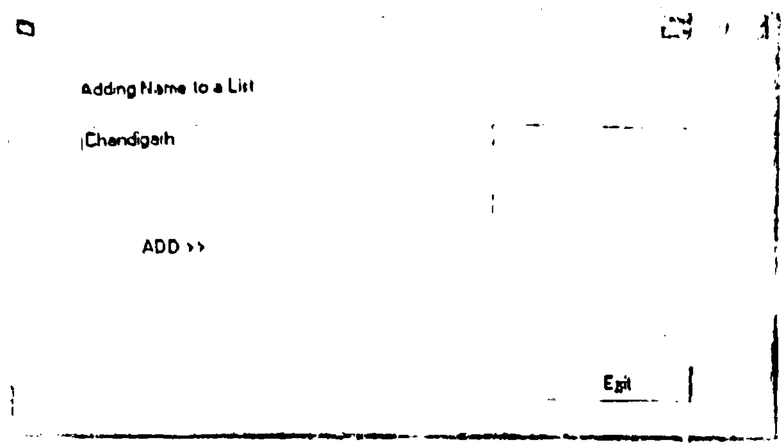
(ii) All questions are compulsory.

(iii) Instructions on how to attempt a question are mentioned against it.

(iv) Total marks assigned to each question are **twenty**.

Q1	**(10X2=20 Marks)**	
(a)	Discuss the internal architecture of JVM with help of a diagram.	**CO1**
(b)	List & explain the features of Java.	
(c)	Discuss the importance of string constant pool. Also, compare StringBuffer with String classes.	
Q2	**(10X2=20 Marks)**	
(a)	What do you mean by run time polymorphism? How "***one interface multiple method***" form of polymorphism can be achieved. Explain with suitable programming code.	**CO2**
(b)	Differentiate between the following: 1) Abstract Class vs Interface 2) Overloading and Overriding	
(c)	Design a class to represent bank account. Includes the following members: • Name of depositor • Account number	

	• Type of account • Balance amount in the account Methods: • To assign initial values • To deposit an amount • To withdraw an amount after checking balance. • To display the name and balance. Write a program to incorporate the constructor to provide initial values, use **this** keyword and also instantiate its object.	
Q3	**(10X2=20 Marks)**	
(a)	Write a program to create two threads, one thread will print odd numbers and second thread will print even numbers between 1 to 20 numbers.	**CO3**
(b)	Explain the concept of I/O streams and different streams including character stream, byte stream and file stream. Explain the concept of I/O streams and different streams including character stream, byte stream and file stream.	
(c)	Define an exception called "MyException" that is thrown when a string is not equal to "India" or "INDIA". Write a program that uses this exception.	
Q 4	**(10X2=20 Marks)**	
(a)	Write a program to create GUI that accepts a city name from the user (through textbox) and adds it to a list box by clicking on the add button. Also write an event handler code for **add>>** and **Exit** buttons. Its screenshot is shown below:	**CO4**

		
(b)	Write the differences between AWT & Swing in Java? Write suitable code to create two text fields to take two integer inputs a,b and a button (submit) in a frame. When the button gets clicked the $(a^2 - b^2)$ operation will be performed and the result will be shown in the frame.	
(c)	What is Collection and Generic Framework in Java? Write java code to insert 10, 20 at the end and 30 at index 2 of an ArrayList<Integer> having elements 1,2,3,4,5.	
Q 5	**(10X2=20 Marks)**	
(a)	Explain the following: i) Event Delegation Model ii) Adapter Classes	
(b)	What are JDBC drivers in Java? Write about at least two such drivers with suitable diagrams with their unique roles in database connectivity.	CO5
(c)	Describe the various steps for using JDBC? Write a program to demonstrate each step with suitable example.	

Roll No.						

Even End Sem. Examination 2022

Name of the course: B.Tech

Semester: II

Name of the Paper: Engineering Chemistry

Paper Code: TCH-201

Maximum Marks:100

Time: 3 Hour's

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b and c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(10 X 2= 20 Marks)	
(a)	Draw the molecular orbital diagram of O_2 molecule and also discuss the bond order & magnetic behavior of this molecule.	CO 1
(b)	What is hydrogen bonding? Differentiate between Intermolecular and Intramolecular hydrogen bonding.	
(c)	Write a note on (i) Principle and application of Spectroscopy. (ii) Nano Scale Materials	
Q2	**(10 X 2= 20 Marks)**	
(a)	Discuss the zeolite method for softening of water with the help of appropriate reactions.	CO 2
(b)	A water sample on analysis give the following data: $Mg(HCO_3)_2$=16.2 mg/L; $Ca(HCO_3)_2$=7.8 mg/L; $CaSO_4$=9.4 mg/L; $MgCl_2$= 17.4mg/L ; $CaCl_2$=11.1 mg/L. Calculate the temporary and permanent hardness of water.	
(c)	Write detail note on followings: (i) Hardness in terms of $CaCO_3$Equivalents. (ii) Reverse Osmosis method for softening of water.	

Q3	(10 X 2= 20 Marks)	
(a)	What is polymer? Discuss the preparation and applications of Nylon 6,6 andDacron.	CO 3
(b)	What do you meanPlastics? Differentiate between Thermoplastics and Thermosetting polymers.	
(c)	Writenote on followings: (i) Conducting polymer (ii) Biodegradable polymer	

Q4	(10 X 2= 20 Marks)	
(a)	What do you mean by Fuels? Classify the fuels with examples. Also write the characteristics of a good fuel.	CO 4
(b)	Write note on the followings: (i) Working of Bomb calorimeter (ii) Biogas	
(c)	Define Gross and Net Calorific Value of a fuel. The following data is obtained in the bomb calorimeter experiment: Weight offuel=1.09 gm, water equivalent of calorimeter= 550 gm, water taken in the calorimeter= 2200 gm, observed rise in temp.=2.3°C, cooling correction= 0.0047 °C, acid correction= 60 calorie, fuse wire correction= 10 calorie. Calculate the gross and net calorific value of the fuel sample. If the fuel contains 6.5% Hydrogen. (latent heat of condensation=580 cal/gm)	

Q5	(10 X 2= 20 Marks)	
(a)	Define the term Electrode Potential and the factors affecting the electrode potential. Write the significances of Electrochemical series.	

(b)	Calculate the electrode potential of a copper wire dipped in 0.1M Cu SO4 at 25 degree centigrade. The standard electrode potential of $E^{o}_{Cu2+/Cu}$ = +0.34 V.	CO 5
(c)	Write note on the followings: (i) Electrochemical theory of Corrosion (ii) Fuel cells	

Roll No. | | | | | | | |

Paper Code: TCS-201

Even End Sem Back Paper Examination December 2022

Course - B.Tech *Semester* -2^{nd}

Subject Name: Programming for Problem Solving

Subject Code: TCS 201

Time: Three Hours

MM: 100

Note:

i. This question paper contains five questions.

ii. All Questions are compulsory.

iii. Instructions on how to attempt a question are mentioned against it.

iv. Total marks assigned to each questions are twenty.

Q1. (Attempt any two questions of choice from a, b and c). (2*10=20 Marks)

a.

i. Design a C program to accept N real numbers from the user, compute the average and then store the numbers those are above the average in another array called **aboveAvg** and below average in the array **belowAvg**. Display the average and content of both the arrays. Write the corresponding algorithm for the same.

ii. Describe multi-dimensional arrays with an illustration. List the advantages of Arrays in real world applications.

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i. Define an Array. Explain & illustrate the significance of **Row Major Order** method of storage of array elements.

ii. Implement a C program by accepting an integer through the command line arguments and compute its factorial. Display the computed factorial to the output screen.

c. Write a C program to check that inputted string is palindrome or not. (without using string library function)

Q2. (Attempt any two questions of choice from a, b and c). (2*10=20 Marks)

a.

i. Explain the storage of strings in C by emphasizing on the significance of **null character**. Describe the working of **strncmp** and **strncpy** functions with an example.

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a. Define the following terms with an example for each: **2.5 x 4 = 10 M**

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a.

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Enter the quantity: 200

Enter the price per sapling: 50

Enter the name of the sapling: **Hibiscus**

Enter the quantity: 100

Enter the price per sapling: 35

Total Cost of maintaining the Nursery (Rs): **13500**

c. Write a C program to read records from a file and calculate grade of each student and display it.

Grade - A if marks>=75

Grade - B if marks>=60

Grade - C if marks>=50

FAIL if marks>=0

Q5. (Attempt any two questions of choice **from a, b and c**). (2*10=20 Marks)

a.

i. Describe the working of **range** function with a sample code illustrating the same. **3M**

ii. Write a Python code to accept a list of integers from the user and then display the same by sorting it in descending order. **4M**

iii. List the differences between list and a tuple. **3M**

b.

i. Dictionary in Python is more flexible data structure that allows storing a pair of items. Support the statement with an example.

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    printf("%d  %d\n",strlen(str),sizeof(str));
    return 0;
}
```

</td>
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ii)int sum(int); [4]

```
int main()
{
        printf( " %d ", sum(146) );
        printf( " %d ", sum(681) );
        return 0;
}
int sum(int n)
{
        static int s;
        while(n!=0)
        {
                s=s+n%10;
                n=n/10;
        }
        return sum;
}
```

</td>
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<tr>
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iii)int main() [2]

```
{
 enum value={val1=0,val2,val3,val4,val5}var;
 printf("%d",sizeof(var));
 return 0;
}
```

</td>
<td>

iv)int main() [2]

```
{
 struct emp
 {
  char n[20];
  int age;
 };
 struct emp e1={"Dravid",23};
 struct emp e2=e1;
 if(e1==e2)
 printf("The structure variable are equal ");
}
```

</td>
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